

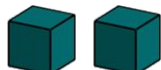


Problem solving with theoretical probability

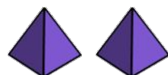
Task A: Using probability to inform decisions

- 1) Andeep is playing a game that involves choosing two dice, rolling them and summing their numbers.

Two of the dice are cubes containing numbers 1 to 6



Two of the dice are tetrahedrons containing numbers 1 to 4



The sample space could be represented with a two-way outcome table.

	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

- a) Which combination of dice give the greatest probability of rolling a total of 8?

$$2 \text{ tetrahedrons: } P(8) = \frac{1}{16} = 0.0625$$

$$1 \text{ of each dice: } P(8) = \frac{3}{24} = 0.125$$

$$2 \text{ cubes: } P(8) = \frac{5}{36} = 0.13\dot{8}$$

2 cube dice

- c) Which combination of dice give the greatest probability of rolling a total of 7?

$$2 \text{ tetrahedrons: } P(7) = \frac{2}{16} = 0.125$$

$$1 \text{ of each dice: } P(7) = \frac{4}{24} = 0.1\dot{6}$$

$$2 \text{ cubes: } P(7) = \frac{6}{36} = 0.1\dot{6}$$

either 2 cube dice or 1 of each dice

- b) Which combination of dice give the greatest probability of rolling a total of 6?

$$2 \text{ tetrahedrons: } P(6) = \frac{3}{16} = 0.1875$$

$$1 \text{ of each dice: } P(6) = \frac{4}{24} = 0.1\dot{6}$$

$$2 \text{ cubes: } P(6) = \frac{5}{36} = 0.13\dot{8}$$

2 tetrahedron dice

- d) Write a summary that advises a player how to decide which pair of dice gives them the greatest probability of rolling the total they want.

Use two tetrahedral dice when you want to roll a total of 6 or less.

Use two cube dice when you want to roll a total of 7 or more.

Alternatively, you could use one of each dice if you want to roll a total of 7.



Task B: Number of trials required to observe a difference

1) Lucas creates two versions of the same video game: Version A and Version B.

In Version A, the probability that a player finds a gem is 0.02

Players are 10% more likely to find a gem in Version B than in Version A.

a) Complete the table.

$$0.02 \times 1.1 = 0.022$$

$$0.022 - 0.02 = 0.002$$

$$0.02 \times 1000 = 20$$

$$0.022 \times 1000 = 22$$

$$22 - 20 = 2$$

	Version A	Version B	difference
probability of finding gem	0.02	0.022	0.002
expected gems per 100 turns	2	2.2	0.2
expected gems per 1000 turns	20	22	2

b) How many turns would it take to expect a difference of 1 gem between the two versions of the game?

500 turns

$$0.022x - 0.02x = 1$$

$$0.002x = 1$$

$$x = 500$$

÷ 2

Alternatively, proportional reasoning could be used.

expected gems per 1000 turns	20	22	2
	↓ ÷ 2	↓ ÷ 2	↓ ÷ 2
expected gems per x turns	10	11	1

2) Sofia creates a video game where players have a 0.5% probability of getting caught in a trap each turn.

She creates a new version of the game where players are now 20% more likely to get caught in a trap each turn, compared to the original version of the game.

a) What is the probability of getting caught in a trap in the new version of the game?

$$0.5\% \times 1.2 = 0.6\%$$

b) How many turns would it take to expect a difference of 1 extra trap between the two versions of the game?

$$0.006x - 0.005x = 1$$

$$0.001x = 1$$

$$x = 1000$$

1000 turns

